

ZENITH RAI

Machine Learning & Backend Engineer

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PROFILE

BSc. CSIT student at Tribhuvan University with hands-on experience building end-to-end machine learning systems and backend architectures. Comfortable implementing ML algorithms from scratch (gradient descent) and serving them behind authenticated REST APIs backed by relational databases. Seeking an internship to apply Python, FastAPI, and ML frameworks on a real-world engineering team.

EDUCATION

Tribhuvan University (TU) | Kathmandu, Nepal

BSc. Computer Science & Information Technology (CSIT) — In Progress, expected 2026 December

+2 / Higher Secondary Education GPA: 3.18 / 4.00

Secondary Education Examination (SEE) GPA: 3.80 / 4.00

TECHNICAL SKILLS

Backend & APIs: FastAPI, PostgreSQL, Railway, SQL(raw)

Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib, Supervised/Unsupervised Learning

Deep Learning & AI: PyTorch, TensorFlow & Keras, CNNs, Local LLMs (Ollama)

Frontend: HTML, CSS, JavaScript

Languages: Python, SQL, C++

PROJECTS

Full-Stack Car Resell Price Prediction API | *FastAPI, PostgreSQL, Scikit-learn, NumPy*

- Built a multi-linear regression model from scratch using mini-batch stochastic gradient descent to predict used car resale prices, reaching 0.87 R².
- Engineered a RESTful FastAPI backend to serve real-time model predictions.
- Integrated a PostgreSQL database to store and retrieve user prediction history.

Website with Attendance System | *HTML, CSS, JavaScript, FastAPI (Railway), PostgreSQL*

- Designed and built a full-stack website with a manual attendance check-in form, backed by a FastAPI REST API and PostgreSQL database deployed on Railway.
- Implemented secure login for staff/admin access and an admin dashboard to view and review submitted attendance records.
- Built the frontend from scratch with HTML, CSS, and JavaScript, handling form validation and API communication with the backend.

Household Inventory & Document Management System | *HTML, CSS, JavaScript, FastAPI (Railway), PostgreSQL*

- Built a personal full-stack web app to catalog household inventory and store key documents in a centralized, searchable system.
- Developed a FastAPI backend with authenticated login, deployed on Railway, backed by a PostgreSQL database for structured record storage.
- Designed a responsive HTML/CSS/JavaScript frontend for adding, editing, and retrieving inventory and document records.

Local RAG Document Q&A System | *Ollama, LLMs, Python*

- Developed a local Retrieval-Augmented Generation (RAG) pipeline using Ollama as the LLM backend, enabling fully offline natural-language Q&A over PDF documents.
- Implemented document chunking, embedding, and vector retrieval to ground answers in source text and preserve data privacy without relying on external APIs.

Multi-Class Eye Disease Detection | *CNN, PyTorch*

- Built a Convolutional Neural Network to classify eye scans into three conditions (diabetic retinopathy, cataract, glaucoma) from labeled medical image data, reaching 96% test accuracy.
- Applied image preprocessing and multi-layer feature extraction & transfer learning to improve classification accuracy across all classes.

Heart Disease Classifier | *Logistic Regression, Scikit-learn*

- Developed a logistic regression classifier to predict heart disease likelihood from patient health indicators, achieving 98% accuracy.
- Executed end-to-end data preprocessing and feature selection pipelines to optimize model performance.